

597 Elevated and Stable Arousal and Wakefulness Regulation in Major Depressive Disorder: Positive Association with Depression Severity

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Background: An upregulated brain arousal has been suggested as a possible pathogenetic factor for major depressive disorder (MDD), as shown by resting-state EEG studies and clinical symptomatology such as high inner tension and long sleep-onset latencies. The main aims of this study are to investigate i) whether depression severity is associated with an upregulation of arousal and wakefulness regulation (AWR) and ii) whether changes in AWR are associated with changes in depressive severity.

Methods: Data from 564 inpatients with moderate-to-severe MDD (MADRS ≥ 20) enrolled in the multicenter naturalistic P4D study across five German university hospitals were analyzed at baseline (day 1–7) and follow-up (day 49 $\pm$ 3). Self-reported AWR was assessed using the Arousal and Wakefulness Regulation Scale (AWRS), a scale measuring behavioral correlates of neurobiological process governing AWR, while depression severity was determined using the Hamilton Depression Scale (HDRS-24). Partial Spearman correlations were computed, controlling for age and gender. Change scores were defined as the difference between follow-up and baseline.

Results: At baseline, elevated and stable AWR, as indexed by higher AWRS scores, significantly correlated with elevated depressive severity ($n = 564$; $p = 0.32$, $p < .001$). Reduction in AWRS scores significantly correlated with reduction in depressive severity over treatment ($n = 319$; $p = 0.40$, $p < .001$).

Discussion: These findings suggest that elevated and stable AWR is an important and state dependent factor in the pathogenesis of depression. Comparison of euthymic people with a

lifetime diagnosis of MDD to healthy controls is necessary to determine whether upregulated AWR is a state-modulated trait.

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